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Newcomers Training

String

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String Definition

- What is the **String** ?

  String is an array of characters and it has some special functions to help.
  The library of the **String** is ***#include<string>***

- What is the **String** functions ?

1- str.size() / str.length()

2- str.push_back()

3- str.pop_back()

4- str.erase()

5- str.substr()

6- str.resize()

7- str.insert()

8- str.back()

9- getline()

String Functions

1- str.size() / str.length()

This function return the size / length of the string, in other word, it return the number of characters that the string has.

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "ahmed";
    cout << s.size() << endl;
    return 0;
}
```

Output : 5

String Functions

2- str.push_back(char C)

This function add the character (**C**) in the end of the string.

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Ahmed";
    s.push_back('F');
    cout << s << endl;
    return 0;
}
```

Output : AhmedF

String Functions

3- str.pop_back()

This function remove the last character from the string.

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "ahmed";
    s.pop_back();
    cout << s << endl;
    return 0;
}
```

Output : ahme

String Functions

4- str.erase(int pos, int len)

This function remove a substring from the string.
This substring start from index (**pos**) and the length of this substring is (**len**)
0 <= pos < str.size(), if (pos) doesn`t satisfy this condition, you will get Runtime Error

str.erase(int pos)
this line will remove a substring, start from index (**pos**) to the end of the string.

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Ibrahim";
    s.erase(1, 4);
    cout << s << endl;
    return 0;
}
```

Output : Iim

String Functions

5- str.substr(int pos, int len)

This function return a substring from the string.
This substring start from index (**pos**) and the length of this substring is (**len**)
0 <= pos < str.size(), if (pos) doesn`t satisfy this condition, you will get Runtime Error

str.substr(int pos)
this line will return a substring, start from index (**pos**) to the end of the string.

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Ibrahim";
    cout << s.substr(2, 5) << endl;
    return 0;
}
```

Output : rahim

String Functions

6- str.resize(int len)

This function change the size of the string to (**len**).

- if (**len**) is smaller than the original size of the string, the string will contain the first (**len**) characters and the remain characters will be removed.

- if (**len**) is greater than the original size of the string, the string add white spaces in the end of it until it`s size will equal (**len**)

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string a = "Mohamed";
    string b = "Ahmed";
    a.resize(2);
    b.resize(7);
    cout << b << a;
    return 0;
}
```

Output : Ahmed Mo

String Functions

7- str.insert(int pos, string S)

This function insert a string or a character to the original string.
This function will start to insert (**S**) in index (**pos**) in the original string and shift right the remaining characters

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string a = "Aed";
    a.insert(1, "hm");
    cout << a;
    return 0;
}
```

Output : Ahmed

String Functions

8- str.back()

This function return the last character in the string

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string s = "Ibrahim";
    cout << s.back() << endl;
    return 0;
}
```

Output : m

String Functions

9- getline(cin, string S)

This function help you to take all the line in the string with the white spaces.

getline(cin, string S, char c)
This line will take all the line until he found the character (**c**) then stop and will not take (**c**).

```
#include<iostream>
#include<string>
using namespace std;
int main(){
    string a;
    getline(cin, a);
    cout << a;
    return 0;
}
```

Input : Ahmed Mohamed Ali

Output : Ahmed Mohamed Ali

For more information about **string** visit this Link

Now it's time to practise and solve the problems of string

String Sheet

Good luck <3